Lab 2 — Common Issues

ECON 470 — Spring 2026

A handful of patterns came up across many of your submissions. None of them is fatal on its own, but together they account for most of the points the class lost. Read these alongside the answer key.

Problem 1, Q2–Q3: posteriors are *not* e and $1 - e$

This was the single most common slip. Three groups reported

$$\Pr(\theta = 1 \mid s = 1) = 0.85 \quad \text{and} \quad \Pr(\theta = 1 \mid s = 2) = 0.15,$$

i.e., they read the accuracy off as the posterior. The accuracy is $\Pr(s = 1 \mid \theta = 1) = e = 0.85$ — that's a *conditional on the state*. The posterior conditions the other way: on the *signal*. Bayes's rule:

$$\Pr(\theta = 1 \mid s = 1) = \frac{\Pr(\theta = 1, s = 1)}{\Pr(s = 1)} = \frac{0.595}{0.595 + 0.045} = \frac{0.595}{0.640} \approx 0.9297.$$

A useful sanity check: if the posterior really were just e, the prior would be irrelevant — and you'd update to the same belief from a prior of 0.7 as from a prior of 0.01. That can't be right. The posterior depends on both the prior *and* the signal accuracy; that is the whole content of Bayes's rule.

A second sanity check: the two posteriors $\Pr(\theta = 1 \mid s = 1) \approx 0.9297$ and $\Pr(\theta = 1 \mid s = 2) \approx 0.2917$ should bracket the prior $p = 0.7$. The signal pulls the posterior up if it's positive and down if it's negative — but it cannot pull *both* posteriors above the prior or both below it.

Problem 2, Q14: a demand curve is a graph of $D(P)$, not just two endpoints

The demand curve runs from $(P, D) = (1.5, 1000)$ to $(30, 0)$. Most graphs got the endpoints right but a few had axis problems:

- **Wrong intercepts.** One graph showed an intercept at $D \approx 1052.6$ (this is $1000 \cdot 200/190$ — what you get if you plug $P = 0$ into $D(P) = 1000(200 - P/0.15)/190$ outside its domain of validity). The piecewise definition matters: $D(P) = 1000$ for $P < 1.5$, and the formula only applies on $[1.5, 30]$. Don't extrapolate the linear segment past its endpoints.
- **Both intercepts unlabeled.** "Demand curve" without numbers labeling where it hits the axes is hard to read and impossible to compare against the answer key. Always label both intercepts.

The cleanest graphs in Q14 set up two clearly-labeled axes (P horizontal, D vertical), drew the segment from $(1.5, 1000)$ to $(30, 0)$, and added a flat top at $D = 1000$ for $P < 1.5$ and a flat bottom at $D = 0$ for $P > 30$.

Problem 2, Q17: it's not "the curve gets bowed"; it's that *every* WTP rises

A few answers said better prediction "shifts the curve out" or "makes the curve more bowed." That language is fine in an aggregate-demand-and-supply context, but Q17 asks for the *economic mechanism*. The mechanism is one line:

$$\text{WTP}_i = (e - p)(R_i - r_i),$$

so WTP_i is increasing in e — for **every** physician, not just some. A higher e raises the entire demand curve because every individual's willingness to pay rose; equivalently, the cutoff $R_i - r_i \geq P/(e - p)$ that determines who adopts is lower at every P, so more physicians clear the bar. This is the per-individual analogue of the lecture's "value of information" formula, applied one decision-maker at a time.

Problem 3, Q21–Q22: the optimum is $(e_1^*, e_2^*) = (1, 0.4)$, not $(0.7, 0.7)$

A few graphs treated the constraint $\frac{e_1+e_2}{2} = 0.70$ as if it forced $e_1 = e_2 = 0.7$. It doesn't — it's a *line*, and the agent can pick any point on it. The interesting feasible points are the *corners* of the segment, $(e_2, e_1) = (0.40, 1.00)$ and $(1.00, 0.40)$, because the objective is linear and the optimum lies at one of them.

Which corner? Compute the indifference-curve slope versus the constraint slope:

- Constraint slope (in (e_2, e_1) space, with e_1 on the vertical axis): -1.
- Indifference-curve slope: $-0.025/0.475 = -1/19$.

The indifference curves are *much flatter* than the constraint, so the highest indifference line touches the constraint at the **upper-left** corner: $(e_2^*, e_1^*) = (0.40, 1.00)$. The agent prioritizes sensitivity (e_1).

The intuition: $p = 0.95$ means 95% of the joint mass sits in the $\theta = 1$ row. Accuracy in that row (the coefficient on e_1) is worth 19 times more than accuracy in the $\theta = 2$ row (the coefficient on e_2). When one input is 19× more valuable than the other, you spend everything on the valuable one and accept the constraint floor on the other.

Problem 3, Q23: "incremental cost relative to the default" is the whole point

The default action (no signal) is $a = 1$ — mark everything as not-spam. Q23 asks how much *extra* cost a misclassification imposes *relative to that default*:

- **False positive** (spam gets through; $\theta = 2$, $s = 1$, agent picks $a = 1$): the default would have done the same thing. The signal didn't change the action. **Incremental cost: 0.**
- **False negative** (legitimate email flagged as spam; $\theta = 1$, $s = 2$, agent switches to $a = 2$): the default would have gotten this right. The signal *caused* the wrong action. **Incremental cost:** $R - r = 0.5$**.**

The asymmetry — only one of the two error types imposes an incremental cost — is *why* the agent sets $e_1 = 1$. False negatives are eliminated entirely; false positives are tolerated because they are free relative to the default. Most submissions got the right numbers; the ones that lost points stated the costs without connecting them back to the default-action benchmark, so the asymmetry didn't have anywhere to land.

Problem 3, Q25: "shifts up and to the left" is loose; the endpoints move *along the axes*

The accuracy frontier in ROC space is the segment from $(\text{FPR}, \text{TPR}) = (0, 2\bar{\lambda}-1)$ to $(2-2\bar{\lambda}, 1)$. When $\bar{\lambda}$ rises:

- The lower endpoint $(0, 2\bar{\lambda} - 1)$ slides **up** the vertical axis (still on $\text{FPR} = 0$).
- The upper endpoint $(2 - 2\bar{\lambda}, 1)$ slides **left** along the top of the unit square (still on $\text{TPR} = 1$).

So the segment as a whole moves toward the corner $(0, 1)$ — but neither endpoint is moving "up and to the left" simultaneously. Drawing it that way (as if the whole segment were translating along a diagonal) hides what is actually happening. The cleanest version of this answer says explicitly that the *frontier collapses to the point* $(0, 1)$ *in the limit* $\bar{\lambda} \to 1$.

Problem 4, Q27: the L-shape is the *limit* of the frontier, not the feasible set

When $\bar{\lambda} = 1$ exactly, the constraint $e_1 + e_2 = 2$ with $e_i \in [0, 1]$ forces $e_1 = e_2 = 1$. The feasible set at $\bar{\lambda} = 1$ is just the single point $(0, 1)$. Gans's green L-shape isn't the feasible set at $\bar{\lambda} = 1$; it's the *limit of the accuracy frontier* as $\bar{\lambda} \to 1$, traced out by the two endpoint trajectories from Q25. That's a small distinction but worth getting right because the L-shape is *not* a menu of options the agent can pick from — it represents an idealization that no real prediction technology achieves.

A note on showing work

Two patterns made grading harder than it needed to be:

1. **Bare numbers in P1 and P2.** Reporting $D(12) = 632$ without showing $1000 \cdot (200 - 80)/190$ is technically correct but gives no partial credit if the final number is off. Show the substitution.
2. **Graphs without axis labels or marked optima.** A graph in Q24 without FPR and TPR labeled (or without the optimal point marked) is asking the grader to guess what you intended. The correct answer has a *specific* point on it: $(0.6, 1)$. Mark it.

The two cleanest submissions in the class did both of these consistently — every numerical answer had its derivation, and every graph had labeled axes, labeled intercepts, and a marked optimum.